

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:58:40

Source: data/json/noise_experiments/v2/noise2i_full8match_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_full8match/noise2i_full8match_game_000_neutral_rep00.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_full8match
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 0
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$3; payoffs Agent_1=\$6, Agent_2=\$3;

noise/visible: sent visible=\$1.86, received visible=\$5.58, returned visible=\$3.06

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 6.0, "min": 0, "raw": 3.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 6.0,

"Agent_2": 3.0

}

Visible balances after round: {

"Agent_1": 6.0600000000000005,

"Agent_2": 2.58

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

```
'''json
{"send": 2}
'''
```

Reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get \$5 - \$2 + \$3 = \$6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
```

```

"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Assistant output:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: \$5 - \$1.86 + \$3.00 = \$6.14 (profit of \$1.14)
- I end with: \$5.58 - \$3.00 = \$2.58 (profit of \$2.58)

This rewards their trust and encourages cooperation.

```
{'return': 3.00}
```

Reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is $\$1.86 \times 3$)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: $\$5 - \$1.86 + \$0 = \3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more
 - So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives..

```
..
Usage: {
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$1.5; payoffs Agent_2=\$4.5, Agent_1=\$4.5;

```

noise/visible: sent visible=$0.9, received visible=$2.7, returned visible=$1.82
Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 6.0, "min": 0, "raw": 1.5}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
  "Agent_1": 10.5,
  "Agent_2": 7.5
}
Visible balances after round: {
  "Agent_1": 7.2600000000000001,
  "Agent_2": 7.4
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: $\$5 - \$1.86 + \$3.00 = \6.14 (profit of \$1.14)
- I end with: $\$5.58 - \$3.00 = \$2.58$ (profit of \$2.58)

This rewards their trust and encourages cooperation.

`{'return': 3.00}`

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is $\$1.86 \times 3$)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: $\$5 - \$1.86 + \$0 = \3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more

- So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

Reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- \$5 - \$2 + \$3 = \$6 (profit of \$1)

This seems reasonable and builds cooperation.

```
Usage: {
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

```
'''json
{"send": 2}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get \$5 - \$2 + \$3 = \$6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Assistant output:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning approximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

Reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2×3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: $\$5 - \$2 = \$3$
- But my total is \$6.06

So the receiver must have returned: $\$6.06 - \$3 = \$3.06$

That's interesting - the receiver returned \$3.06 out of the \$6.00 they received. That's about 51% - a relatively fair split.

Now I'm in the receiver position. The sender sent me \$0.9 (which is less than the \$2 I sent them in Round 1). I received \$2.7.

I should consider:

1. The sender showed some trust by sending \$0.9
2. In Round 1, when I sent \$2, they returned about 51% (\$3.06 out of \$6)
3. I want to maintain a cooperative relationship
4. I should return a fair amount to encourage future cooperation

If I return about half of what I received: $\$2.7 / 2 = \1.35

This would give the sender:

- They started with \$5
- They sent \$0.9, so had \$4.1
- Plus my return of \$1.35 = \$5.45 total
- Net gain: \$0.45

And I would keep:

- $\$2.7 - \$1.35 = \$1.35$

This seems fair - we both profit equally from the exchange.

```
...
Usage: {
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2.5; received \$7.5; Agent_2 returned \$0; payoffs Agent_1=\$2.5, Agent_2=\$7.5;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 13.0,

"Agent_2": 15.0

}

Visible balances after round: {

"Agent_1": 9.760000000000002,

"Agent_2": 7.4

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 3,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not over exposing myself.

```
'''json
{"send": 2}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than

if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get $\$5 - \$2 + \$3 = \6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning a pproximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2×3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: $\$5 - \$2 = \$3$
- But my total is \$6.06

So the receiver must have returned: $\$6.06 - \$3 = \$3.06$

That's interesting - the receiver returned \$3.06 out of the \$6.00 they received. That's about 51% - a relatively fair split.

Now I'm in the receiver position. The sender sent me \$0.9 (which is less than the \$2 I sent them in Round 1). I received \$2.7.

I should consider:

1. The sender showed some trust by sending \$0.9
2. In Round 1, when I sent \$2, they returned about 51% (\$3.06 out of \$6)
3. I want to maintain a cooperative relationship
4. I should return a fair amount to encourage future cooperation

If I return about half of what I received: $\$2.7 / 2 = \1.35

- They started with \$5
- They sent \$0.9, so had \$4.1
- Plus my return of \$1.35 = \$5.45 total
- Net gain: \$0.45

$$- \$2.7 - \$1.35 = \$1.35$$

• • •

```
{
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
```

Round 3

How much do you send to the receiver this round? (0-5)

Round 1: I sent \$2 and received a good return (~\$3)
Round 2: As receiver, I got \$0.9 (which became \$2.7) and returned \$1.5

```
'''json
{"send": 2.5}
'''
```

Round 1: I was the SENDER

- I sent \$2
- This became \$6 to the receiver
- From my total earnings of \$6.06, this suggests I received back about \$3.06
- I kept: $\$5 - \$2 + \$3.06 = \6.06 ?

Round 3: I'm the SENDER again
- Total earnings so far: \$7.26
- I have a new \$5 to work with

1. They're being more cautious
2. There's communication noise (mentioned in instructions)
3. It's a different strategy

I'll send \$2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.

```
Usage: {  
  "input_tokens": 713,  
  "output_tokens": 547,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:56:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: $\$5 - \$1.86 + \$3.00 = \6.14 (profit of \$1.14)
- I end with: $\$5.58 - \$3.00 = \$2.58$ (profit of \$2.58)

This rewards their trust and encourages cooperation.

{'return': 3.00}

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is $\$1.86 \times 3$)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: $\$5 - \$1.86 + \$0 = \3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$

2. Return enough so sender breaks even plus a bit more:

- Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
- Or return more to reward trust

3. Return enough so we both gain equally from the transaction:

- The multiplication created \$3.72 in value ($\$5.58 - \1.86)
- Splitting this gain equally means each of us gets \$1.86 more
- So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- \$5 - \$2 + \$3 = \$6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

Usage: {

```
"input_tokens": 838,
"output_tokens": 207,
"reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1.5; received \$4.5; Agent_1 returned \$0; payoffs Agent_2=\$3.5, Agent_1=\$4.5;
noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

```
"Agent_1": 17.5,
"Agent_2": 18.5
}
```

Visible balances after round: {

```
"Agent_1": 9.760000000000002,
"Agent_2": 10.9
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:56:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment

nt. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: $\$5 - \$1.86 + \$3.00 = \6.14 (profit of \$1.14)
- I end with: $\$5.58 - \$3.00 = \$2.58$ (profit of \$2.58)

This rewards their trust and encourages cooperation.

{'return': 3.00}

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is $\$1.86 \times 3$)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: $\$5 - \$1.86 + \$0 = \3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more
 - So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money.

A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- \$5 - \$2 + \$3 = \$6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, these decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less than Round 2 to reflect the reduced trust level.

```
{'send': 1.50}
```

Reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: $(\$5 - \$2.00 + \text{returned}) + \$2.58 = \$7.40$
- $\$3.00 + \text{returned} + \$2.58 = \$7.40$
- returned = \$1.82

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining
- ...

```
Usage: {  
  "input_tokens": 931,  
  "output_tokens": 714,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:56:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
  Looking at this first round, I need to balance trust-building with caution since I don't h
  ave any history with this receiver yet.

  Sending some amount can create mutual benefit - if I send money, it triples, and the recei
  ver can return enough for both of us to profit. However, I'm taking a risk that they mig
  ht keep everything.

  For Round 1, I'll send a moderate amount to signal willingness to cooperate while not over
  exposing myself.

  '''json
  {"send": 2}
  '''
reasoning:
  This is the first round of a trust game. I'm the sender with $5.

  Let me think about the optimal strategy:
  - If I send $X, the receiver gets $3X
  - The receiver then decides how much to return
  - I keep: $5 - X + returned amount
  - Receiver keeps: 3X - returned amount

  The game theory aspect here is that this involves trust. If I send money, I'm trusting the
  receiver to return some amount. The rational thing for the receiver would be to return
  enough to incentivize continued cooperation if this is a repeated game.

```

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get $\$5 - \$2 + \$3 = \6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning approximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2×3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: $\$5 - \$2 = \$3$
- But my total is \$6.06

- I have a new \$5 to work with

Looking at the pattern:

- In Round 1, I sent \$2 and got back \$3.06 (good return!)
- In Round 2, the sender sent me \$0.9 (less than what I sent in Round 1)
- I returned \$1.5 out of \$2.7 (about 55%)

The sender in Round 2 sent less than I did in Round 1. This could be:

1. They're being more cautious
2. There's communication noise (mentioned in instructions)
3. It's a different strategy

Since I received a good return in Round 1, and I want to maintain cooperation, I should send a reasonable amount. However, the Round 2 sender was more conservative, sending only \$0.9.

I'll send a moderate amount - not as high as Round 1 since cooperation seems mixed, but enough to show I'm still willing to cooperate.

I'll send \$2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.

usage:

```
{
  "input_tokens": 713,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
'''json
{"return": 0}
'''
```

Usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$1; received \$3; Agent_2 returned \$1; payoffs Agent_1=\$5, Agent_2=\$2;

noise/visible: sent visible=\$0.65, received visible=\$1.95, returned visible=\$1.09

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 3.0, "min": 0, "raw": 1.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 22.5,

"Agent_2": 20.5

}

Visible balances after round: {

"Agent_1": 14.850000000000001,

"Agent_2": 11.850000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:56:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 5,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not over exposing myself.

```
'''json
{"send": 2}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get \$5 - \$2 + \$3 = \$6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning a pproximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2 × 3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: \$5 - \$2 = \$3
- But my total is \$6.06

So the receiver must have returned: \$6.06 - \$3 = \$3.06

That's interesting - the receiver returned \$3.06 out of the \$6.00 they received. That's about 51% - a relatively fair split.

Now I'm in the receiver position. The sender sent me \$0.9 (which is less than the \$2 I sent them in Round 1). I received \$2.7.

I should consider:

1. The sender showed some trust by sending \$0.9
2. In Round 1, when I sent \$2, they returned about 51% (\$3.06 out of \$6)
3. I want to maintain a cooperative relationship
4. I should return a fair amount to encourage future cooperation

If I return about half of what I received: $\$2.7 / 2 = \1.35

This would give the sender:

- They started with \$5
- They sent \$0.9, so had \$4.1
- Plus my return of \$1.35 = \$5.45 total
- Net gain: \$0.45

And I would keep:

- $\$2.7 - \$1.35 = \$1.35$

• • •

```
{
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
```

Round 3

How much do you send to the receiver this round? (0-5)

Looking at the history:

```
'''json
{"send": 2.5}
'''
```

Let me track what's happened so far:

```
{
  "input_tokens": 713,
  "output_tokens": 547,
```

```

    "reasoning_tokens": 0
  }
Message 8 role=user
  content:
    Round 4

    You are the RECEIVER this round. Your total earnings across all rounds are $9.760000000000000002.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 9 role=assistant
  content:
    Looking at this round:

    - The sender sent me $0, so I received $0
    - There's nothing for me to return since I didn't receive anything

    This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

    Since I received $0, I can only return:

    '''json
    {"return": 0}
    '''
  reasoning:
    The sender sent me $0, so I received $0. Since there's nothing to return, I can only return $0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

    However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

    Since I received $0, I can only return between $0 and $0.

    '''json
    {"return": 0}
    '''
  usage:
    {
      "input_tokens": 879,
      "output_tokens": 248,
      "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 5

    You are the SENDER this round. Your total earnings across all rounds are $9.760000000000000002.
    How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at the recent pattern:

  Round 3: I sent $2.5 and my earnings only increased by $2.5 (meaning I got $0 back)
  Round 4: As receiver, I got $0 from the sender

  Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

  I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

  '''json
  {"send": 1}
  '''
  Reasoning:

```

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)
Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5
Round 3: I was SENDER, sent \$2.5, but then...
Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26
- In Round 3, I was SENDER and sent \$2.5
- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5
- I kept \$2.5
- I should have received something back
- My total went from \$7.26 to \$9.76, an increase of \$2.5
- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)
- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)
- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

```
Usage: {  
  "input_tokens": 1029,  
  "output_tokens": 655,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:56:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: \$5 - \$1.86 + \$3.00 = \$6.14 (profit of \$1.14)
- I end with: \$5.58 - \$3.00 = \$2.58 (profit of \$2.58)

This rewards their trust and encourages cooperation.

{'return': 3.00}

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is \$1.86 × 3)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: $\$5 - \$1.86 + \$0 = \3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more
 - So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- $\$5 - \$2 + \$3 = \6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, these decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less than Round 2 to reflect the reduced trust level.

```
{'send': 1.50}
```

reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: ($\$5 - \$2.00 + \text{returned}$) + $\$2.58 = \7.40
- $\$3.00 + \text{returned} + \$2.58 = \$7.40$
- $\text{returned} = \$1.82$

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of

what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining

...

usage:

```
{
  "input_tokens": 931,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.9.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Assistant output:

Looking at this round:

- The sender sent \$0.65 (13% of endowment) - showing limited trust
- I received \$1.95

The cooperation level has been declining, but I want to maintain some reciprocity to potential ly rebuild trust. The sender did show some willingness to cooperate, even if minimal.

I'll return approximately half of what I received, similar to my approach in Round 1. This ensures both of us gain something:

- Sender would get: \$5 - \$0.65 + \$1.00 = \$5.35
- I would get: \$1.95 - \$1.00 = \$0.95

This rewards their small gesture of trust while we both profit modestly.

{'return': 1.00}

Reasoning:

Let me analyze this situation:

The sender sent \$0.65, which is 13% of their endowment. This is quite low compared to previous rounds where I've seen 37.2% and I sent 40%.

I received \$1.95 (the multiplication by 3).

Looking at the pattern:

- Round 1: I received \$5.58, returned \$3.00 (54%)
- Round 2: I sent \$2.00 (40%), but got low returns
- Round 3: Sender sent \$0, I returned \$0
- Round 4: I sent \$1.50
- Round 5: Now I received \$1.95 (sender sent \$0.65)

The cooperation has been declining. The sender only sent 13% of their endowment this round, showing limited trust.

However, I should consider:

1. There's communication noise - maybe they intended to send more
2. Reciprocity can rebuild trust
3. Being too punishing could end cooperation entirely

A fair approach would be to return roughly half of what I received, similar to my Round 1 strategy. This would give the sender a small profit for their trust while I also gain.

If I return \$1.00:

- Sender ends with: \$5 - \$0.65 + \$1.00 = \$5.35 (profit of \$0.35)
- I end with: \$1.95 - \$1.00 = \$0.95

This maintains some reciprocity while acknowledging the lower initial trust.

Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown, but still be cooperative.

Let me return \$0.90 (about 46% of what I received):

...

```
Usage: {
  "input_tokens": 1136,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1.75; received \$5.25; Agent_1 returned \$0.7; payoffs Agent_2=\$3.95, Agent_1=\$4.55; noise/visible: sent visible=\$0.46, received visible=\$1.38, returned visible=\$0
Action validation: {"returned": {"amount": 0.7, "clamped": false, "max": 5.25, "min": 0, "raw": 0.7}, "sent": {"amount": 1.75, "clamped": false, "max": 5, "min": 0, "raw": 1.75}}

Balances after round: {

"Agent_1": 27.05,
"Agent_2": 24.45

}

Visible balances after round: {

"Agent_1": 15.530000000000001,
"Agent_2": 15.100000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:57:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: \$5 - \$1.86 + \$3.00 = \$6.14 (profit of \$1.14)
- I end with: \$5.58 - \$3.00 = \$2.58 (profit of \$2.58)

This rewards their trust and encourages cooperation.

{'return': 3.00}

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is 1.86×3)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: \$5 - \$1.86 + \$0 = \$3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: $\$5 - \$1.86 + \$5.58 = \8.72
- I get: $\$0$

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more
 - So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- $\$5 - \$2 + \$3 = \6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, these decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less than Round 2 to reflect the reduced trust level.

```
{'send': 1.50}
```

reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: ($\$5 - \$2.00 + \text{returned}$) + $\$2.58 = \7.40
- $\$3.00 + \text{returned} + \$2.58 = \$7.40$
- returned = \$1.82

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:
- There's been some cooperation but declining

...

usage:

```
{
  "input_tokens": 931,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.9.
This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 11 role=assistant

content:

Looking at this round:

- The sender sent \$0.65 (13% of endowment) - showing limited trust
- I received \$1.95

The cooperation level has been declining, but I want to maintain some reciprocity to potentially rebuild trust. The sender did show some willingness to cooperate, even if minimal.

I'll return approximately half of what I received, similar to my approach in Round 1. This ensures both of us gain something:

- Sender would get: $\$5 - \$0.65 + \$1.00 = \5.35
- I would get: $\$1.95 - \$1.00 = \$0.95$

This rewards their small gesture of trust while we both profit modestly.

{'return': 1.00}

reasoning:

Let me analyze this situation:

The sender sent \$0.65, which is 13% of their endowment. This is quite low compared to previous rounds where I've seen 37.2% and I sent 40%.

I received \$1.95 (the multiplication by 3).

Looking at the pattern:

- Round 1: I received \$5.58, returned \$3.00 (54%)
- Round 2: I sent \$2.00 (40%), but got low returns
- Round 3: Sender sent \$0, I returned \$0
- Round 4: I sent \$1.50
- Round 5: Now I received \$1.95 (sender sent \$0.65)

The cooperation has been declining. The sender only sent 13% of their endowment this round, showing limited trust.

However, I should consider:

1. There's communication noise - maybe they intended to send more
2. Reciprocity can rebuild trust
3. Being too punishing could end cooperation entirely

A fair approach would be to return roughly half of what I received, similar to my Round 1 strategy. This would give the sender a small profit for their trust while I also gain.

If I return \$1.00:

- Sender ends with: $\$5 - \$0.65 + \$1.00 = \5.35 (profit of \$0.35)
- I end with: $\$1.95 - \$1.00 = \$0.95$

This maintains some reciprocity while acknowledging the lower initial trust.

Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown, but still be cooperative.

Let me return \$0.90 (about 46% of what I received):

...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.8500000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience:

****As Receiver:**** I've been fairly reciprocal (Round 1: returned \$3/\$5.58; Round 5: returned \$1/\$1.95)

****As Sender:****

- Round 2: Sent \$2.00, received \$1.82 back (decent reciprocity)
- Round 4: Sent \$1.50, received \$0 back (no reciprocity)

There's been inconsistency, possibly due to communication noise mentioned in the instructions. However, in Round 5, I showed fair reciprocity, which might encourage cooperation.

I'll send a moderate amount that balances trust with caution:

{'send': 1.75}

Reasoning:

Let me review the rounds so far:

Round 1: I was receiver, sender sent \$1.86, I received \$5.58, I returned \$3.00

Round 2: I was sender, sent \$2.00, receiver returned some amount (I ended with \$7.4 total, so started with \$2.58, sent \$2.00, got \$5 + return - \$2 = \$3 + return. So \$7.4 - \$2.58 = \$4.82 gained. Started with \$5, sent \$2, so I kept \$3 + return. Return must have been \$1.82)

Round 3: I was receiver, sender sent \$0, I returned \$0. My total stayed at \$7.4

Round 4: I was sender, sent \$1.50, started with \$7.4, so I have \$5 from this round, sent \$1.50, so kept \$3.50 + return. Total is now \$10.9. So \$10.9 - \$7.4 = \$3.5, which means return was \$0. Hmm.

Round 5: I was receiver, sender sent \$0.65, I received \$1.95, I returned \$1.00. Started with \$10.9, gained \$0.95, so total is \$11.85.

So the pattern:

- Round 1: I received and reciprocated well
- Round 2: I sent \$2.00, got back \$1.82 (reasonable)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.50, got back \$0 (no reciprocation!)
- Round 5: I received \$0.65*3=\$1.95, returned \$1.00 (fair reciprocation)

So when I was receiver (Rounds 1, 3, 5), I was fairly generous.

When I was sender (Rounds 2, 4):

- Round 2 ...

Usage: {

```
"input_tokens": 1360,
"output_tokens": 837,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:57:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
```

```

"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.


```
'''json
{"send": 2}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets 3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get \$5 - \$2 + \$3 = \$6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning approximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

'''

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2×3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: $\$5 - \$2 = \$3$
- But my total is \$6.06

So the receiver must have returned: $\$6.06 - \$3 = \$3.06$

That's interesting - the receiver returned \$3.06 out of the \$6.00 they received. That's about 51% - a relatively fair split.

Now I'm in the receiver position. The sender sent me \$0.9 (which is less than the \$2 I sent them in Round 1). I received \$2.7.

I should consider:

1. The sender showed some trust by sending \$0.9
2. In Round 1, when I sent \$2, they returned about 51% (\$3.06 out of \$6)
3. I want to maintain a cooperative relationship
4. I should return a fair amount to encourage future cooperation

If I return about half of what I received: $\$2.7 / 2 = \1.35

This would give the sender:

- They started with \$5
- They sent \$0.9, so had \$4.1
- Plus my return of \$1.35 = \$5.45 total
- Net gain: \$0.45

And I would keep:

- $\$2.7 - \$1.35 = \$1.35$

This seems fair - we both profit equally from the exchange.

...

usage:

```
{
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2600000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the history:

Round 1: I sent \$2 and received a good return (~\$3)

Round 2: As receiver, I got \$0.9 (which became \$2.7) and returned \$1.5

The cooperation has been decent so far. Both players have shown willingness to share gains. I'll increase my send slightly to continue building trust and maximize mutual benefit.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me track what's happened so far:

Round 1: I was the SENDER

- I sent \$2
- This became \$6 to the receiver
- From my total earnings of \$6.06, this suggests I received back about \$3.06
- I kept: $\$5 - \$2 + \$3.06 = \6.06 ?

Round 2: I was the RECEIVER

- Sender sent me \$0.9, which became \$2.7
- I returned \$1.5
- I kept: $\$2.7 - \$1.5 = \$1.2$
- Total so far: $\$6.06 + \$1.2 = \$7.26$?

Round 3: I'm the SENDER again

- Total earnings so far: \$7.26
- I have a new \$5 to work with

Looking at the pattern:

- In Round 1, I sent \$2 and got back \$3.06 (good return!)
- In Round 2, the sender sent me \$0.9 (less than what I sent in Round 1)
- I returned \$1.5 out of \$2.7 (about 55%)

The sender in Round 2 sent less than I did in Round 1. This could be:

1. They're being more cautious
2. There's communication noise (mentioned in instructions)
3. It's a different strategy

Since I received a good return in Round 1, and I want to maintain cooperation, I should send a reasonable amount. However, the Round 2 sender was more conservative, sending only \$0.9.

I'll send a moderate amount - not as high as Round 1 since cooperation seems mixed, but enough to show I'm still willing to cooperate.

I'll send \$2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.

usage:

```
{
  "input_tokens": 713,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to

```
send anything this round.
```

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.76000000000000

2.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$2.5 and my earnings only increased by \$2.5 (meaning I got \$0 back)

Round 4: As receiver, I got \$0 from the sender

Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)

Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5

Round 3: I was SENDER, sent \$2.5, but then...

Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26
- In Round 3, I was SENDER and sent \$2.5
- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5
- I kept \$2.5
- I should have received something back
- My total went from \$7.26 to \$9.76, an increase of \$2.5
- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)
- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)

- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

usage:

```
{
  "input_tokens": 1029,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.850000000000001.

This round, The sender sent you \$0.46, so you received \$1.3800000000000001. How much do you return to The sender? (0-1.3800000000000001)

Assistant output:

Looking at this round:

- The sender sent \$0.46, which became \$1.38 for me
- This is a small send, possibly testing cooperation after recent issues
- I want to encourage rebuilding trust by being fair

I'll return a reasonable amount that allows both of us to profit from this exchange:

```
'''json
{"return": 0.7}
'''
```

Reasoning:

Let me analyze this situation:

Round 5: I sent \$1 as a sender (presumably to this receiver)

Round 6: As receiver, I'm now getting \$0.46 which became \$1.38

The sender is sending a relatively small amount (\$0.46), possibly testing the waters after the breakdown in cooperation.

Looking at my earnings jump from \$9.76 to \$14.85, that's an increase of about \$5.09. That suggests in Round 5, I received a decent return - maybe around \$5 back after sending \$1.

Now the sender is being cautious with only \$0.46. I should return a fair amount to encourage rebuilding trust. Since they sent \$0.46 and I received \$1.38, I should return enough that they profit slightly.

If I return around \$0.7-\$0.9, the sender would:

- Started with \$5
- Sent \$0.46, has \$4.54
- Gets back ~\$0.7-\$0.9
- Ends with ~\$5.24-\$5.44 (profit of \$0.24-\$0.44)

I would keep:

- \$1.38 - \$0.8 = \$0.58 profit

Let me return about \$0.7 to be fair and signal cooperation.

Usage: {

```
"input_tokens": 1225,
"output_tokens": 394,
"reasoning_tokens": 0
```

}

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 30.05,

"Agent_2": 30.45

}

Visible balances after round: {

"Agent_1": 18.53,

"Agent_2": 15.100000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:57:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with caution since I don't have any history with this receiver yet.

Sending some amount can create mutual benefit - if I send money, it triples, and the receiver can return enough for both of us to profit. However, I'm taking a risk that they might keep everything.

For Round 1, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

```
'''json
{"send": 2}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return
- I keep: \$5 - X + returned amount
- Receiver keeps: 3X - returned amount

The game theory aspect here is that this involves trust. If I send money, I'm trusting the receiver to return some amount. The rational thing for the receiver would be to return enough to incentivize continued cooperation if this is a repeated game.

Since this is Round 1, I don't have any history with this receiver. However, the instructions mention "multiple rounds," so this suggests repeated interaction.

A common strategy in trust games:

- Sending \$0 is safe but yields no potential gain
- Sending some amount shows trust and can lead to mutual benefit
- Sending all \$5 maximizes the total pot but is very risky

A moderate approach would be to send around \$2-3 to show some trust while not risking everything. This allows:

- Testing the receiver's trustworthiness
- Creating opportunity for mutual gain
- Not losing everything if the receiver doesn't reciprocate

Let me send \$3. This creates a pot of \$9 for the receiver. If they return even half (\$4.50), I'd end up with \$6.50 (\$5 - \$3 + \$4.50), and they'd keep \$4.50. Both better off than if I sent nothing.

Actually, let me reconsider. A \$2 send might be more conservative for the first round:

- Receiver gets \$6
- If they return \$3, I get \$5 - \$2 + \$3 = \$6
- They keep \$3
- Both benefit

I'll go with \$2 for the first round as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

I'll return a fair amount that allows both of us to profit from this exchange. Returning a pproximately half seems reasonable and reciprocal.

```
'''json
{"return": 1.5}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

My total earnings so far are \$6.06. Let me think about what happened in Round 1:

- I was the SENDER and sent \$2
- The receiver got \$6 (2 × 3)
- My earnings are \$6.06

If I kept everything from being the sender:

- I would have: \$5 - \$2 = \$3
- But my total is \$6.06

So the receiver must have returned: \$6.06 - \$3 = \$3.06

That's interesting - the receiver returned \$3.06 out of the \$6.00 they received. That's about 51% - a relatively fair split.

Now I'm in the receiver position. The sender sent me \$0.9 (which is less than the \$2 I sent them in Round 1). I received \$2.7.

I should consider:

1. The sender showed some trust by sending \$0.9
2. In Round 1, when I sent \$2, they returned about 51% (\$3.06 out of \$6)
3. I want to maintain a cooperative relationship
4. I should return a fair amount to encourage future cooperation

If I return about half of what I received: $2.7 / 2 = 1.35$

This would give the sender:

- They started with \$5
- They sent \$0.9, so had \$4.1
- Plus my return of \$1.35 = \$5.45 total
- Net gain: \$0.45

And I would keep:

- $2.7 - 1.35 = 1.35$

This seems fair - we both profit equally from the exchange.


```

...
usage:
{
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
Message 6 role=user
content:
  Round 3

  You are the SENDER this round. Your total earnings across all rounds are $7.260000000000000
  1.
  How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
content:
  Looking at the history:

  Round 1: I sent $2 and received a good return (~$3)
  Round 2: As receiver, I got $0.9 (which became $2.7) and returned $1.5

  The cooperation has been decent so far. Both players have shown willingness to share gains
  . I'll increase my send slightly to continue building trust and maximize mutual benefit.

  '''json
  {"send": 2.5}
  '''
reasoning:
  Let me track what's happened so far:

  Round 1: I was the SENDER
  - I sent $2
  - This became $6 to the receiver
  - From my total earnings of $6.06, this suggests I received back about $3.06
  - I kept: $5 - $2 + $3.06 = $6.06 ?

  Round 2: I was the RECEIVER
  - Sender sent me $0.9, which became $2.7
  - I returned $1.5
  - I kept: $2.7 - $1.5 = $1.2
  - Total so far: $6.06 + $1.2 = $7.26 ?

  Round 3: I'm the SENDER again
  - Total earnings so far: $7.26
  - I have a new $5 to work with

  Looking at the pattern:
  - In Round 1, I sent $2 and got back $3.06 (good return!)
  - In Round 2, the sender sent me $0.9 (less than what I sent in Round 1)
  - I returned $1.5 out of $2.7 (about 55%)

  The sender in Round 2 sent less than I did in Round 1. This could be:
  1. They're being more cautious
  2. There's communication noise (mentioned in instructions)
  3. It's a different strategy

  Since I received a good return in Round 1, and I want to maintain cooperation, I should send a reasonable amount. However, the Round 2 sender was more conservative, sending only $0.9.

  I'll send a moderate amount - not as high as Round 1 since cooperation seems mixed, but enough to show I'm still willing to cooperate.

  I'll send $2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.
usage:
{
  "input_tokens": 713,
  "output_tokens": 547,
  "reasoning_tokens": 0
}

```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.760000000000000002.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$2.5 and my earnings only increased by \$2.5 (meaning I got \$0 back)
Round 4: As receiver, I got \$0 from the sender

Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)
Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5
Round 3: I was SENDER, sent \$2.5, but then...
Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26
- In Round 3, I was SENDER and sent \$2.5
- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5
- I kept \$2.5
- I should have received something back
- My total went from \$7.26 to \$9.76, an increase of \$2.5
- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)
- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)
- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

usage:

```
{
  "input_tokens": 1029,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.850000000000001.

This round, The sender sent you \$0.46, so you received \$1.3800000000000001. How much do you return to The sender? (0-1.3800000000000001)

Message 13 role=assistant

content:

Looking at this round:

- The sender sent \$0.46, which became \$1.38 for me
- This is a small send, possibly testing cooperation after recent issues
- I want to encourage rebuilding trust by being fair

I'll return a reasonable amount that allows both of us to profit from this exchange:

```
'''json
{"return": 0.7}
'''
```

reasoning:

Let me analyze this situation:

Round 5: I sent \$1 as a sender (presumably to this receiver)

Round 6: As receiver, I'm now getting \$0.46 which became \$1.38

The sender is sending a relatively small amount (\$0.46), possibly testing the waters after

the breakdown in cooperation.

Looking at my earnings jump from \$9.76 to \$14.85, that's an increase of about \$5.09. That suggests in Round 5, I received a decent return - maybe around \$5 back after sending \$1.

Now the sender is being cautious with only \$0.46. I should return a fair amount to encourage rebuilding trust. Since they sent \$0.46 and I received \$1.38, I should return enough that they profit slightly.

If I return around \$0.7-\$0.9, the sender would:

- Started with \$5
- Sent \$0.46, has \$4.54
- Gets back ~\$0.7-\$0.9
- Ends with ~\$5.24-\$5.44 (profit of \$0.24-\$0.44)

I would keep:

- \$1.38 - \$0.8 = \$0.58 profit

Let me return about \$0.7 to be fair and signal cooperation.

usage:

```
{
  "input_tokens": 1225,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.5300000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at recent trends:

Round 5: I sent \$1 and got back about \$1.09 (modest but positive)

Round 6: As receiver, I got \$1.38 (from \$0.46 sent) and returned \$0.7 fairly

Cooperation seems to be rebuilding after the breakdown in Rounds 3-4. I'll send a moderate amount to encourage this positive trend while being cautious:

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me track the history:

Round 1 (SENDER): I sent \$2

Round 2 (RECEIVER): I received \$2.7 (from \$0.9 sent), returned \$1.5

Round 3 (SENDER): I sent \$2.5

Round 4 (RECEIVER): I received \$0 (from \$0 sent), returned \$0

Round 5 (SENDER): I sent \$1

Round 6 (RECEIVER): I received \$1.38 (from \$0.46 sent), returned \$0.7

My total earnings are \$15.53.

Let me trace what I should have earned:

- Round 1: Started with \$5, sent \$2, kept \$3. Need to know what was returned.
- Round 2: As receiver, got \$2.7, returned \$1.5, kept \$1.2
- Round 3: Started with \$5, sent \$2.5, kept \$2.5. Need to know what was returned.
- Round 4: As receiver, got \$0, returned \$0, kept \$0
- Round 5: Started with \$5, sent \$1, kept \$4. Need to know what was returned.
- Round 6: As receiver, got \$1.38, returned \$0.7, kept \$0.68

Let me work backwards from \$15.53:

- After Round 6: earned \$0.68, so before Round 6 I had \$15.53 - \$0.68 = \$14.85 (matches what was stated)
- After Round 5: Let's say I got back X from Round 5. Before Round 5 I had \$14.85 - \$4 - X = \$10.85 - X
- After Round 4: earned \$0, so before Round 4 I had \$10.85 - X

- After Round 3: If I got back Y from Round 3, before Round 3 I had $\$10.85 - X - \$2.5 - Y = \$8.35 - X - Y$
- After Round 2: earned \$1.2, so before Round 2 I had $\$7.15 - X - Y$
- After Round 1: If I got back Z from Round 1, before Round 1 I had $\$4.15 - X - Y - Z$

Since I started with \$0, the returns must sum to \$4.15: $X + Y + Z = \$4.15$

From the pattern of each round:

- Round 1: Sent \$2 which became \$6, earnings were \$6.06, so $Z = \$3.06$
- Round 3: Sent \$2.5 which became \$7.5, earnings were \$9.76, and I had \$7.26 before this round, meaning $Y = 0$
- Round 5: Sent \$1 which became \$3, and I had \$14.85 after this round

, so $X = \$0.41$

Let me recalculate systematically. Round 1 as sender: started with \$5, sent \$2, kept \$3, ended with \$6.06, so received back \$3.06. Round 2 as receiver: got \$2.7, returned \$1.5, kept \$1.2, total became \$7.26. Round 3 as sender: started with \$5, sent \$2.5, kept \$2.5, ended with \$9.76, so received back \$0. Round 4 as receiver: received \$0.

In Round 5 as sender, I started with \$5, sent \$1, kept \$4. Before Round 6 I had \$14.85, and I kept \$0.68 in that round, so after Round 5 I must have had \$14.85, meaning I received back \$1.09. Round 6 as receiver: received \$1.38, returned \$0.7, kept \$0.68, bringing my total to \$15.53....

```
Usage: {
  "input_tokens": 1358,
  "output_tokens": 1594,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:57:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$1.86, that is 37.2% of their total endowment. You received \$5.58. How much do you return to The sender? (0-5.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$1.86 (37.2% of their endowment)
- This was multiplied to \$5.58 for me
- I want to reciprocate their trust while both of us benefit

The sender took a risk to create mutual gain. A fair approach would be to split the received amount roughly equally, ensuring the sender profits from their trust while I also gain.

If I return about half of what I received:

- Sender ends with: \$5 - \$1.86 + \$3.00 = \$6.14 (profit of \$1.14)
- I end with: \$5.58 - \$3.00 = \$2.58 (profit of \$2.58)

This rewards their trust and encourages cooperation.

{'return': 3.00}

reasoning:

In this round, I am the RECEIVER. Let me analyze the situation:

- The sender had \$5
- The sender sent \$1.86 (which is 37.2% of their \$5 endowment)
- I received \$5.58 (which is \$1.86 × 3)
- I need to decide how much to return (between \$0 and \$5.58)

Let me think about a fair and rational approach:

The sender showed some trust by sending \$1.86. They could have sent nothing, but they chose to send about 37% of their endowment.

If I return nothing:

- Sender gets: \$5 - \$1.86 + \$0 = \$3.14
- I get: \$5.58

If I return everything (\$5.58):

- Sender gets: \$5 - \$1.86 + \$5.58 = \$8.72
- I get: \$0

A common approach in trust games is to try to be fair and reciprocate trust. One way to think about this:

- The sender gave up \$1.86 to give me an opportunity to gain
- I received \$5.58
- A fair split might be to ensure both parties benefit

Some possible strategies:

1. Equal split of the multiplied amount: Return \$2.79 (half of \$5.58)
 - Sender gets: $\$3.14 + \$2.79 = \$5.93$
 - I get: $\$5.58 - \$2.79 = \$2.79$
2. Return enough so sender breaks even plus a bit more:
 - Sender spent \$1.86, so return \$1.86 to break even: Sender gets \$5
 - Or return more to reward trust
3. Return enough so we both gain equally from the transaction:
 - The multiplication created \$3.72 in value ($\$5.58 - \1.86)
 - Splitting this gain equally means each of us gets \$1.86 more
 - So I should return \$3.72, giving the sender \$6.86 and leaving me with \$1.86

But wait, let me verify this differently. If we split the final \$5.58 equally instead, each person gets \$2.79. That would leave the sender with \$5.93 after their initial \$1.86 investment, and me with \$2.79 from my original \$5.

Actually, splitting the profit makes more sense-the \$3.72 gain gets divided equally, so I return the original \$1.86 plus my half of the profit (\$1.86), totaling \$3.72 back. This way we each end up \$1.86 ahead from the transaction.

I should also factor in that the sender took a real risk by trusting me with their money. A reasonable return would be somewhere between \$2.79 and \$3.00, which acknowledges their trust while still letting me benefit from facilitating the transaction.

Given that I started with nothing and the sender started with \$5, returning exactly half of what I received (\$2.79) feels like a fair reciprocal gesture that balances both perspectives....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1833,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

```
{'send': 2.00}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- $\$5 - \$2 + \$3 = \6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

`{'return': 0}`

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
```



```
"input_tokens": 838,  
"output_tokens": 207,  
"reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)
Round 2: I sent \$2.00, but the receiver returned less generously (~30%)
Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, the
se decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less
than Round 2 to reflect the reduced trust level.

```
{'send': 1.50}
```

reasoning:

Let me review what happened:

Round 1: I was RECEIVER
- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER
- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: ($\$5 - \$2.00 + \text{returned}$) + $\$2.58 = \7.40
- $\$3.00 + \text{returned} + \$2.58 = \$7.40$
- $\text{returned} = \$1.82$

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER
- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now
- My total earnings are \$7.40

The pattern so far:
- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of
what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining

...

usage:

```
{  
  "input_tokens": 931,
```

```

    "output_tokens": 714,
    "reasoning_tokens": 0
  }
Message 10 role=user
  content:
    Round 5

    You are the RECEIVER this round. Your total earnings across all rounds are $10.9.
    This round, The sender sent you $0.65, so you received $1.9500000000000002. How much do yo
    u return to The sender? (0-1.9500000000000002)
Message 11 role=assistant
  content:
    Looking at this round:

    - The sender sent $0.65 (13% of endowment) - showing limited trust
    - I received $1.95

    The cooperation level has been declining, but I want to maintain some reciprocity to poten
    tially rebuild trust. The sender did show some willingness to cooperate, even if minimal
    .

    I'll return approximately half of what I received, similar to my approach in Round 1. This
    ensures both of us gain something:

    - Sender would get: $5 - $0.65 + $1.00 = $5.35
    - I would get: $1.95 - $1.00 = $0.95

    This rewards their small gesture of trust while we both profit modestly.

    {'return': 1.00}
  reasoning:
    Let me analyze this situation:

    The sender sent $0.65, which is 13% of their endowment. This is quite low compared to prev
    ious rounds where I've seen 37.2% and I sent 40%.

    I received $1.95 (the multiplication by 3).

    Looking at the pattern:
    - Round 1: I received $5.58, returned $3.00 (54%)
    - Round 2: I sent $2.00 (40%), but got low returns
    - Round 3: Sender sent $0, I returned $0
    - Round 4: I sent $1.50
    - Round 5: Now I received $1.95 (sender sent $0.65)

    The cooperation has been declining. The sender only sent 13% of their endowment this round
    , showing limited trust.

    However, I should consider:
    1. There's communication noise - maybe they intended to send more
    2. Reciprocity can rebuild trust
    3. Being too punishing could end cooperation entirely

    A fair approach would be to return roughly half of what I received, similar to my Round 1
    strategy. This would give the sender a small profit for their trust while I also gain.

    If I return $1.00:
    - Sender ends with: $5 - $0.65 + $1.00 = $5.35 (profit of $0.35)
    - I end with: $1.95 - $1.00 = $0.95

    This maintains some reciprocity while acknowledging the lower initial trust.

    Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown,
    but still be cooperative.

    Let me return $0.90 (about 46% of what I received):
    ...
  usage:
    {
      "input_tokens": 1136,

```

```
    "output_tokens": 656,  
    "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.850000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my experience:

****As Receiver:**** I've been fairly reciprocal (Round 1: returned \$3/\$5.58; Round 5: returned \$1/\$1.95)

****As Sender:****

- Round 2: Sent \$2.00, received \$1.82 back (decent reciprocity)
- Round 4: Sent \$1.50, received \$0 back (no reciprocity)

There's been inconsistency, possibly due to communication noise mentioned in the instructions. However, in Round 5, I showed fair reciprocity, which might encourage cooperation.

I'll send a moderate amount that balances trust with caution:

```
{'send': 1.75}
```

reasoning:

Let me review the rounds so far:

Round 1: I was receiver, sender sent \$1.86, I received \$5.58, I returned \$3.00

Round 2: I was sender, sent \$2.00, receiver returned some amount (I ended with \$7.4 total, so started with \$2.58, sent \$2.00, got \$5 + return - \$2 = \$3 + return. So \$7.4 - \$2.58 = \$4.82 gained. Started with \$5, sent \$2, so I kept \$3 + return. Return must have been \$1.82)

Round 3: I was receiver, sender sent \$0, I returned \$0. My total stayed at \$7.4

Round 4: I was sender, sent \$1.50, started with \$7.4, so I have \$5 from this round, sent \$1.50, so kept \$3.50 + return. Total is now \$10.9. So \$10.9 - \$7.4 = \$3.5, which means return was \$0. Hmm.

Round 5: I was receiver, sender sent \$0.65, I received \$1.95, I returned \$1.00. Started with \$10.9, gained \$0.95, so total is \$11.85.

So the pattern:

- Round 1: I received and reciprocated well
- Round 2: I sent \$2.00, got back \$1.82 (reasonable)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.50, got back \$0 (no reciprocation!)
- Round 5: I received \$0.65*3=\$1.95, returned \$1.00 (fair reciprocation)

So when I was receiver (Rounds 1, 3, 5), I was fairly generous.

When I was sender (Rounds 2, 4):

- Round 2 ...

usage:

```
{  
  "input_tokens": 1360,  
  "output_tokens": 837,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.100000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is disappointing after I sent a moderate amount in the previous round, but I have no choice

ce.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is disappointing as I sent \$1.75 in Round 6, hoping to rebuild cooperation. But if the sender sent \$0, there's nothing I can do.

Usage: {

"input_tokens": 1585,

"output_tokens": 114,

"reasoning_tokens": 0

}

=====

ROUND 8

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0.75; received \$2.25; Agent_1 returned \$1; payoffs Agent_2=\$5.25,

Agent_1=\$1.25; noise/visible: sent visible=\$0.82, received visible=\$2.46, returned visible=\$1.15

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 2.25, "min": 0, "raw": 1.0}, "sent": {"amount": 0.75, "clamped": false, "max": 5, "min": 0, "raw": 0.75}}

Balances after round: {

"Agent_1": 31.3,

"Agent_2": 35.7

}

Visible balances after round: {

"Agent_1": 19.990000000000002,

"Agent_2": 20.5

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:57:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$2.58.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, I experienced a cooperative exchange where trust was reciprocated. The sender sent me 37.2% of their endowment, and I returned a fair amount that benefited both of us.

I'll apply a similar strategy of moderate trust. Sending a reasonable amount can encourage reciprocity and mutual benefit.

I'll send \$2.00 (40% of my endowment), which is slightly more than what was sent to me in Round 1. This demonstrates trust while not being overly risky.

If the receiver reciprocates fairly (as I did), we should both profit from this cooperation.

{'send': 2.00}

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$1.86 (37.2% of their endowment)
- I received \$5.58
- I returned \$3.00
- My earnings were \$2.58

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver decides what to return

The previous sender showed moderate trust by sending 37.2% of their endowment. I reciprocated fairly by returning about half of what I received, which gave them a profit.

Now I should consider:

1. Building trust and cooperation - sending a reasonable amount
2. The receiver might reciprocate similarly to how I did
3. If I send around the same percentage (37.2%), that would be about \$1.86

However, I could also:

- Send more to show higher trust and potentially get more back
- Send less to be more conservative

Given that I showed good reciprocity in Round 1 by returning \$3.00 out of \$5.58 (about 54%), the new receiver might reciprocate similarly if I show trust.

Let me send a similar amount to what was sent to me - around \$1.86 to \$2.00. This shows consistency and moderate trust.

I'll send \$2.00, which is 40% of my endowment. This is slightly more than what was sent to me, showing I'm willing to trust and cooperate.

If the receiver returns about half (similar to what I did), I'd get back around \$3.00, netting me:

- $\$5 - \$2 + \$3 = \6 (profit of \$1)

This seems reasonable and builds cooperation.

usage:

```
{
  "input_tokens": 632,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, the
se decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less
s than Round 2 to reflect the reduced trust level.

{'send': 1.50}

reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58

- I returned \$3.00

- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)

- My total earnings after round 2 are \$7.40

- So I started Round 2 with \$2.58 and ended with \$7.40

- That means: $(\$5 - \$2.00 + \text{returned}) + \$2.58 = \$7.40$

- $\$3.00 + \text{returned} + \$2.58 = \$7.40$

- returned = \$1.82

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0

- I returned \$0 (couldn't return anything)

- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%

- Round 2: I sent 40%, receiver returned about 30%

- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of
what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect

2. A response to something (maybe they saw the Round 2 receiver's low return?)

3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining

...

usage:

```
{
  "input_tokens": 931,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.9.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do yo

u return to The sender? (0-1.9500000000000002)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent \$0.65 (13% of endowment) - showing limited trust
- I received \$1.95

The cooperation level has been declining, but I want to maintain some reciprocity to potentially rebuild trust. The sender did show some willingness to cooperate, even if minimal.

I'll return approximately half of what I received, similar to my approach in Round 1. This ensures both of us gain something:

- Sender would get: $\$5 - \$0.65 + \$1.00 = \5.35
- I would get: $\$1.95 - \$1.00 = \$0.95$

This rewards their small gesture of trust while we both profit modestly.

{'return': 1.00}

reasoning:

Let me analyze this situation:

The sender sent \$0.65, which is 13% of their endowment. This is quite low compared to previous rounds where I've seen 37.2% and I sent 40%.

I received \$1.95 (the multiplication by 3).

Looking at the pattern:

- Round 1: I received \$5.58, returned \$3.00 (54%)
- Round 2: I sent \$2.00 (40%), but got low returns
- Round 3: Sender sent \$0, I returned \$0
- Round 4: I sent \$1.50
- Round 5: Now I received \$1.95 (sender sent \$0.65)

The cooperation has been declining. The sender only sent 13% of their endowment this round, showing limited trust.

However, I should consider:

1. There's communication noise - maybe they intended to send more
2. Reciprocity can rebuild trust
3. Being too punishing could end cooperation entirely

A fair approach would be to return roughly half of what I received, similar to my Round 1 strategy. This would give the sender a small profit for their trust while I also gain.

If I return \$1.00:

- Sender ends with: $\$5 - \$0.65 + \$1.00 = \5.35 (profit of \$0.35)
- I end with: $\$1.95 - \$1.00 = \$0.95$

This maintains some reciprocity while acknowledging the lower initial trust.

Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown, but still be cooperative.

Let me return \$0.90 (about 46% of what I received):

...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.8500000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my experience:

****As Receiver:**** I've been fairly reciprocal (Round 1: returned \$3/\$5.58; Round 5: returned \$1/\$1.95)

****As Sender:****

- Round 2: Sent \$2.00, received \$1.82 back (decent reciprocity)
- Round 4: Sent \$1.50, received \$0 back (no reciprocity)

There's been inconsistency, possibly due to communication noise mentioned in the instructions. However, in Round 5, I showed fair reciprocity, which might encourage cooperation.

I'll send a moderate amount that balances trust with caution:

{'send': 1.75}

reasoning:

Let me review the rounds so far:

Round 1: I was receiver, sender sent \$1.86, I received \$5.58, I returned \$3.00

Round 2: I was sender, sent \$2.00, receiver returned some amount (I ended with \$7.4 total, so started with \$2.58, sent \$2.00, got \$5 + return - \$2 = \$3 + return. So \$7.4 - \$2.58 = \$4.82 gained. Started with \$5, sent \$2, so I kept \$3 + return. Return must have been \$1.82)

Round 3: I was receiver, sender sent \$0, I returned \$0. My total stayed at \$7.4

Round 4: I was sender, sent \$1.50, started with \$7.4, so I have \$5 from this round, sent \$1.50, so kept \$3.50 + return. Total is now \$10.9. So \$10.9 - \$7.4 = \$3.5, which means return was \$0. Hmm.

Round 5: I was receiver, sender sent \$0.65, I received \$1.95, I returned \$1.00. Started with \$10.9, gained \$0.95, so total is \$11.85.

So the pattern:

- Round 1: I received and reciprocated well
- Round 2: I sent \$2.00, got back \$1.82 (reasonable)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.50, got back \$0 (no reciprocation!)
- Round 5: I received \$0.65*3=\$1.95, returned \$1.00 (fair reciprocation)

So when I was receiver (Rounds 1, 3, 5), I was fairly generous.

When I was sender (Rounds 2, 4):

- Round 2 ...

usage:

```
{
  "input_tokens": 1360,
  "output_tokens": 837,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.100000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is disappointing after I sent a moderate amount in the previous round, but I have no choice.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is disappointing as I sent \$1.75 in Round 6, hoping to rebuild cooperation. But if the sender sent \$0, there's nothing I can do.

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.10000000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my sending history:

- Round 2: Sent \$2.00 ? received \$1.82 (modest return)
- Round 4: Sent \$1.50 ? received \$0
- Round 6: Sent \$1.75 ? received \$0

The last two times I've sent money as sender, I received nothing back. This pattern suggests either low reciprocity or significant communication noise affecting the game.

While I've tried to maintain cooperation, the results when I'm the sender have been poor. I need to be more cautious.

I'll send a small amount to signal continued willingness to cooperate without risking much:

```
{'send': 0.75}
```

Reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.00, received \$1.82 back
- Round 4: Sent \$1.50, received \$0 back
- Round 6: Sent \$1.75, received \$0 back (since Round 7 the sender sent me \$0)

As receiver:

- Round 1: Received \$5.58, returned \$3.00
- Round 3: Received \$0, returned \$0
- Round 5: Received \$1.95, returned \$1.00
- Round 7: Received \$0, returned \$0

The pattern shows declining cooperation overall. When I've been the sender recently (Round 4 and 6), I've gotten nothing back. This could be due to communication noise, or it could be that the receivers are not reciprocating.

My total earnings are \$15.10, which is averaging about \$1.89 per round across 8 rounds. This is below the \$5 I'd get by just keeping everything.

Given the pattern of low reciprocity when I'm the sender, I should be more cautious. However, completely stopping cooperation might eliminate any chance of future gains.

I'll send a small amount to test if there's any willingness to cooperate, but protect most of my endowment.

Usage: {

```
"input_tokens": 1431,
"output_tokens": 467,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:58:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
```

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.0600000000000005.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.9, which became \$2.7 after the 3x multiplier
- In Round 1, when I sent \$2 (becoming \$6), the receiver returned about half, which was fair
- I want to maintain a cooperative relationship and encourage future trust

```
'''json
{"return": 1.5}
'''
```

1. I'm now the RECEIVER
2. The sender sent me \$0.9, which was multiplied by 3, so I received \$2.7
3. I need to decide how much to return (0-2.7)

• • •

```
{
  "input_tokens": 549,
  "output_tokens": 678,
  "reasoning_tokens": 0
}
```

. I'll increase my send slightly to continue building trust and maximize mutual benefit.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me track what's happened so far:

Round 1: I was the SENDER

- I sent \$2
- This became \$6 to the receiver
- From my total earnings of \$6.06, this suggests I received back about \$3.06
- I kept: $\$5 - \$2 + \$3.06 = \6.06 ?

Round 2: I was the RECEIVER

- Sender sent me \$0.9, which became \$2.7
- I returned \$1.5
- I kept: $\$2.7 - \$1.5 = \$1.2$
- Total so far: $\$6.06 + \$1.2 = \$7.26$?

Round 3: I'm the SENDER again

- Total earnings so far: \$7.26
- I have a new \$5 to work with

Looking at the pattern:

- In Round 1, I sent \$2 and got back \$3.06 (good return!)
- In Round 2, the sender sent me \$0.9 (less than what I sent in Round 1)
- I returned \$1.5 out of \$2.7 (about 55%)

The sender in Round 2 sent less than I did in Round 1. This could be:

1. They're being more cautious
2. There's communication noise (mentioned in instructions)
3. It's a different strategy

Since I received a good return in Round 1, and I want to maintain cooperation, I should send a reasonable amount. However, the Round 2 sender was more conservative, sending only \$0.9.

I'll send a moderate amount - not as high as Round 1 since cooperation seems mixed, but enough to show I'm still willing to cooperate.

I'll send \$2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.

usage:

```
{
  "input_tokens": 713,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
```

```
    {"return": 0}
    """
```

reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
    """json
    {"return": 0}
    """
```

usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.7600000000000002.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$2.5 and my earnings only increased by \$2.5 (meaning I got \$0 back)

Round 4: As receiver, I got \$0 from the sender

Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

```
    """json
    {"send": 1}
    """
```

reasoning:

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)

Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5

Round 3: I was SENDER, sent \$2.5, but then...

Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26

- In Round 3, I was SENDER and sent \$2.5

- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5

- I kept \$2.5

- I should have received something back

- My total went from \$7.26 to \$9.76, an increase of \$2.5

- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)

- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)
- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

usage:

```
{
  "input_tokens": 1029,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.850000000000001.

This round, The sender sent you \$0.46, so you received \$1.3800000000000001. How much do you return to The sender? (0-1.3800000000000001)

Message 11 role=assistant

content:

Looking at this round:

- The sender sent \$0.46, which became \$1.38 for me
- This is a small send, possibly testing cooperation after recent issues
- I want to encourage rebuilding trust by being fair

I'll return a reasonable amount that allows both of us to profit from this exchange:

```
'''json
{"return": 0.7}
'''
```

reasoning:

Let me analyze this situation:

Round 5: I sent \$1 as a sender (presumably to this receiver)

Round 6: As receiver, I'm now getting \$0.46 which became \$1.38

The sender is sending a relatively small amount (\$0.46), possibly testing the waters after the breakdown in cooperation.

Looking at my earnings jump from \$9.76 to \$14.85, that's an increase of about \$5.09. That suggests in Round 5, I received a decent return - maybe around \$5 back after sending \$1.

Now the sender is being cautious with only \$0.46. I should return a fair amount to encourage rebuilding trust. Since they sent \$0.46 and I received \$1.38, I should return enough that they profit slightly.

If I return around \$0.7-\$0.9, the sender would:

- Started with \$5
- Sent \$0.46, has \$4.54
- Gets back ~\$0.7-\$0.9
- Ends with ~\$5.24-\$5.44 (profit of \$0.24-\$0.44)

I would keep:

- \$1.38 - \$0.8 = \$0.58 profit

Let me return about \$0.7 to be fair and signal cooperation.

usage:

```
{
  "input_tokens": 1225,
```

```
"output_tokens": 394,
"reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.5300000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at recent trends:

Round 5: I sent \$1 and got back about \$1.09 (modest but positive)

Round 6: As receiver, I got \$1.38 (from \$0.46 sent) and returned \$0.7 fairly

Cooperation seems to be rebuilding after the breakdown in Rounds 3-4. I'll send a moderate amount to encourage this positive trend while being cautious:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track the history:

Round 1 (SENDER): I sent \$2

Round 2 (RECEIVER): I received \$2.7 (from \$0.9 sent), returned \$1.5

Round 3 (SENDER): I sent \$2.5

Round 4 (RECEIVER): I received \$0 (from \$0 sent), returned \$0

Round 5 (SENDER): I sent \$1

Round 6 (RECEIVER): I received \$1.38 (from \$0.46 sent), returned \$0.7

My total earnings are \$15.53.

Let me trace what I should have earned:

- Round 1: Started with \$5, sent \$2, kept \$3. Need to know what was returned.
- Round 2: As receiver, got \$2.7, returned \$1.5, kept \$1.2
- Round 3: Started with \$5, sent \$2.5, kept \$2.5. Need to know what was returned.
- Round 4: As receiver, got \$0, returned \$0, kept \$0
- Round 5: Started with \$5, sent \$1, kept \$4. Need to know what was returned.
- Round 6: As receiver, got \$1.38, returned \$0.7, kept \$0.68

Let me work backwards from \$15.53:

- After Round 6: earned \$0.68, so before Round 6 I had $\$15.53 - \$0.68 = \$14.85$ (matches what was stated)
- After Round 5: Let's say I got back X from Round 5. Before Round 5 I had $\$14.85 - \$4 - X = \$10.85 - X$
- After Round 4: earned \$0, so before Round 4 I had $\$10.85 - X$
- After Round 3: If I got back Y from Round 3, before Round 3 I had $\$10.85 - X - \$2.5 - Y = \$8.35 - X - Y$
- After Round 2: earned \$1.2, so before Round 2 I had $\$7.15 - X - Y$
- After Round 1: If I got back Z from Round 1, before Round 1 I had $\$4.15 - X - Y - Z$

Since I started with \$0, the returns must sum to \$4.15: $X + Y + Z = \$4.15$

From the pattern of each round:

- Round 1: Sent \$2 which became \$6, earnings were \$6.06, so $Z = \$3.06$
- Round 3: Sent \$2.5 which became \$7.5, earnings were \$9.76, and I had \$7.26 before this round, meaning $Y = 0$
- Round 5: Sent \$1 which became \$3, and I had \$14.85 after this round

, so $X = \$0.41$

Let me recalculate systematically. Round 1 as sender: started with \$5, sent \$2, kept \$3, ended with \$6.06, so received back \$3.06. Round 2 as receiver: got \$2.7, returned \$1.5, kept \$1.2, total became \$7.26. Round 3 as sender: started with \$5, sent \$2.5, kept \$2.5, ended with \$9.76, so received back \$0. Round 4 as receiver: received \$0.

In Round 5 as sender, I started with \$5, sent \$1, kept \$4. Before Round 6 I had \$14.85, an

d I kept \$0.68 in that round, so after Round 5 I must have had \$14.85, meaning I received back \$1.09. Round 6 as receiver: received \$1.38, returned \$0.7, kept \$0.68, bringing my total to \$15.53....

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 1594,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.53.

This round, The sender sent you \$0.82, so you received \$2.46. How much do you return to The sender? (0-2.46)

Assistant output:

Looking at this situation:

- The sender sent \$0.82, which became \$2.46 for me
- In Round 7, I sent \$2 but seems I got little/nothing back, which is disappointing
- However, continuing a cycle of non-cooperation hurts both of us

I'll return a fair amount to signal willingness to cooperate, but be somewhat cautious given recent history:

```
'''json
{"return": 1}
'''
```

Reasoning:

Looking at the pattern:

Round 7: I sent \$2 and my earnings went from \$15.53 to \$18.53 = \$3 increase

This means: \$5 - \$2 + returned amount = \$3

So: \$3 + returned amount = \$3

Therefore: returned amount ? \$0

So I sent \$2, and apparently got \$0 back (or close to it).

Round 8 (now): As receiver, I got \$0.82 which became \$2.46

This is frustrating. I sent \$2 last round and got nothing back. Now the sender only sends \$0.82.

However, I need to consider:

1. There's communication noise - maybe the sender intended more
2. I committed to being fair and cooperative
3. If I retaliate by returning nothing, we're in a downward spiral

Let me think about what's fair:

- The sender sent \$0.82 out of \$5, keeping \$4.18
- I received \$2.46
- If I return about half (~\$1.2), both of us profit from the exchange
- The sender would get: \$5 - \$0.82 + \$1.2 = \$5.38
- I would get: \$2.46 - \$1.2 = \$1.26

That seems reasonable. But given they returned \$0 last round, maybe I should return less?

Actually, let me recalculate Round 7:

- I started Round 7 at \$15.53
- I sent \$2
- I ended Round 7 at \$18.53
- So I gained \$3

If I sent \$2, I should have: \$5 - \$2 = \$3 remaining, plus whatever was returned.

\$3 + returned = \$3

returned = \$0

Yeah, so they returned nothing to me in Round 7 when I sent \$2.

...

```
Usage: {
  "input_tokens": 1389,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2.5; received \$7.5; Agent_2 returned \$0; payoffs Agent_1=\$2.5, Agent_2=\$7.5;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.9

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 33.8,

"Agent_2": 43.2

}

Visible balances after round: {

"Agent_1": 23.39,

"Agent_2": 20.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.26000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the history:

Round 1: I sent \$2 and received a good return (~\$3)

Round 2: As receiver, I got \$0.9 (which became \$2.7) and returned \$1.5

The cooperation has been decent so far. Both players have shown willingness to share gains . I'll increase my send slightly to continue building trust and maximize mutual benefit.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me track what's happened so far:

Round 1: I was the SENDER

- I sent \$2
- This became \$6 to the receiver
- From my total earnings of \$6.06, this suggests I received back about \$3.06
- I kept: $\$5 - \$2 + \$3.06 = \6.06 ?

Round 2: I was the RECEIVER

- Sender sent me \$0.9, which became \$2.7
- I returned \$1.5
- I kept: $\$2.7 - \$1.5 = \$1.2$
- Total so far: $\$6.06 + \$1.2 = \$7.26$?

Round 3: I'm the SENDER again

- Total earnings so far: \$7.26
- I have a new \$5 to work with

Looking at the pattern:

- In Round 1, I sent \$2 and got back \$3.06 (good return!)
- In Round 2, the sender sent me \$0.9 (less than what I sent in Round 1)
- I returned \$1.5 out of \$2.7 (about 55%)

The sender in Round 2 sent less than I did in Round 1. This could be:

1. They're being more cautious
2. There's communication noise (mentioned in instructions)
3. It's a different strategy

Since I received a good return in Round 1, and I want to maintain cooperation, I should send a reasonable amount. However, the Round 2 sender was more conservative, sending only \$0.9.

I'll send a moderate amount - not as high as Round 1 since cooperation seems mixed, but enough to show I'm still willing to cooperate.

I'll send \$2.5 - a bit more than Round 1 to signal continued trust, but not the maximum.

usage:

```
{
  "input_tokens": 713,
  "output_tokens": 547,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall the instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.760000000000002.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$2.5 and my earnings only increased by \$2.5 (meaning I got \$0 back)
Round 4: As receiver, I got \$0 from the sender

Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)
Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5
Round 3: I was SENDER, sent \$2.5, but then...
Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26
- In Round 3, I was SENDER and sent \$2.5
- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5
- I kept \$2.5
- I should have received something back
- My total went from \$7.26 to \$9.76, an increase of \$2.5
- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)
- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)
- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

usage:

```
{
  "input_tokens": 1029,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.850000000000001.

This round, The sender sent you \$0.46, so you received \$1.3800000000000001. How much do you return to The sender? (0-1.3800000000000001)

Message 9 role=assistant

content:

Looking at this round:

- The sender sent \$0.46, which became \$1.38 for me

- This is a small send, possibly testing cooperation after recent issues
- I want to encourage rebuilding trust by being fair

I'll return a reasonable amount that allows both of us to profit from this exchange:

```
'''json
{"return": 0.7}
'''
```

reasoning:

Let me analyze this situation:

Round 5: I sent \$1 as a sender (presumably to this receiver)

Round 6: As receiver, I'm now getting \$0.46 which became \$1.38

The sender is sending a relatively small amount (\$0.46), possibly testing the waters after the breakdown in cooperation.

Looking at my earnings jump from \$9.76 to \$14.85, that's an increase of about \$5.09. That suggests in Round 5, I received a decent return - maybe around \$5 back after sending \$1.

Now the sender is being cautious with only \$0.46. I should return a fair amount to encourage rebuilding trust. Since they sent \$0.46 and I received \$1.38, I should return enough that they profit slightly.

If I return around \$0.7-\$0.9, the sender would:

- Started with \$5
- Sent \$0.46, has \$4.54
- Gets back ~\$0.7-\$0.9
- Ends with ~\$5.24-\$5.44 (profit of \$0.24-\$0.44)

I would keep:

- \$1.38 - \$0.8 = \$0.58 profit

Let me return about \$0.7 to be fair and signal cooperation.

usage:

```
{
  "input_tokens": 1225,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.530000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at recent trends:

Round 5: I sent \$1 and got back about \$1.09 (modest but positive)

Round 6: As receiver, I got \$1.38 (from \$0.46 sent) and returned \$0.7 fairly

Cooperation seems to be rebuilding after the breakdown in Rounds 3-4. I'll send a moderate amount to encourage this positive trend while being cautious:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track the history:

Round 1 (SENDER): I sent \$2

Round 2 (RECEIVER): I received \$2.7 (from \$0.9 sent), returned \$1.5

Round 3 (SENDER): I sent \$2.5

Round 4 (RECEIVER): I received \$0 (from \$0 sent), returned \$0

Round 5 (SENDER): I sent \$1

Round 6 (RECEIVER): I received \$1.38 (from \$0.46 sent), returned \$0.7

My total earnings are \$15.53.

Let me trace what I should have earned:

- Round 1: Started with \$5, sent \$2, kept \$3. Need to know what was returned.
- Round 2: As receiver, got \$2.7, returned \$1.5, kept \$1.2
- Round 3: Started with \$5, sent \$2.5, kept \$2.5. Need to know what was returned.
- Round 4: As receiver, got \$0, returned \$0, kept \$0
- Round 5: Started with \$5, sent \$1, kept \$4. Need to know what was returned.
- Round 6: As receiver, got \$1.38, returned \$0.7, kept \$0.68

Let me work backwards from \$15.53:

- After Round 6: earned \$0.68, so before Round 6 I had $\$15.53 - \$0.68 = \$14.85$ (matches what was stated)
- After Round 5: Let's say I got back X from Round 5. Before Round 5 I had $\$14.85 - \$4 - X = \$10.85 - X$
- After Round 4: earned \$0, so before Round 4 I had $\$10.85 - X$
- After Round 3: If I got back Y from Round 3, before Round 3 I had $\$10.85 - X - \$2.5 - Y = \$8.35 - X - Y$
- After Round 2: earned \$1.2, so before Round 2 I had $\$7.15 - X - Y$
- After Round 1: If I got back Z from Round 1, before Round 1 I had $\$4.15 - X - Y - Z$

Since I started with \$0, the returns must sum to \$4.15: $X + Y + Z = \$4.15$

From the pattern of each round:

- Round 1: Sent \$2 which became \$6, earnings were \$6.06, so $Z = \$3.06$
- Round 3: Sent \$2.5 which became \$7.5, earnings were \$9.76, and I had \$7.26 before this round, meaning $Y = 0$
- Round 5: Sent \$1 which became \$3, and I had \$14.85 after this round

, so $X = \$0.41$

Let me recalculate systematically. Round 1 as sender: started with \$5, sent \$2, kept \$3, ended with \$6.06, so received back \$3.06. Round 2 as receiver: got \$2.7, returned \$1.5, kept \$1.2, total became \$7.26. Round 3 as sender: started with \$5, sent \$2.5, kept \$2.5, ended with \$9.76, so received back \$0. Round 4 as receiver: received \$0.

In Round 5 as sender, I started with \$5, sent \$1, kept \$4. Before Round 6 I had \$14.85, and I kept \$0.68 in that round, so after Round 5 I must have had \$14.85, meaning I received back \$1.09. Round 6 as receiver: received \$1.38, returned \$0.7, kept \$0.68, bringing my total to \$15.53....

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 1594,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.53.

This round, The sender sent you \$0.82, so you received \$2.46. How much do you return to The sender? (0-2.46)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$0.82, which became \$2.46 for me
- In Round 7, I sent \$2 but seems I got little/nothing back, which is disappointing
- However, continuing a cycle of non-cooperation hurts both of us

I'll return a fair amount to signal willingness to cooperate, but be somewhat cautious given recent history:

```
'''json
{"return": 1}
'''
```

reasoning:

Looking at the pattern:

Round 7: I sent \$2 and my earnings went from \$15.53 to \$18.53 = \$3 increase
This means: $\$5 - \$2 + \text{returned amount} = \3
So: $\$3 + \text{returned amount} = \3
Therefore: returned amount ? \$0

So I sent \$2, and apparently got \$0 back (or close to it).

Round 8 (now): As receiver, I got \$0.82 which became \$2.46

This is frustrating. I sent \$2 last round and got nothing back. Now the sender only sends \$0.82.

However, I need to consider:

1. There's communication noise - maybe the sender intended more
2. I committed to being fair and cooperative
3. If I retaliate by returning nothing, we're in a downward spiral

Let me think about what's fair:

- The sender sent \$0.82 out of \$5, keeping \$4.18
- I received \$2.46
- If I return about half (~\$1.2), both of us profit from the exchange
- The sender would get: $\$5 - \$0.82 + \$1.2 = \5.38
- I would get: $\$2.46 - \$1.2 = \$1.26$

That seems reasonable. But given they returned \$0 last round, maybe I should return less?

Actually, let me recalculate Round 7:

- I started Round 7 at \$15.53
- I sent \$2
- I ended Round 7 at \$18.53
- So I gained \$3

If I sent \$2, I should have: $\$5 - \$2 = \$3$ remaining, plus whatever was returned.
 $\$3 + \text{returned} = \3
returned = \$0

Yeah, so they returned nothing to me in Round 7 when I sent \$2.

...

usage:

```
{
  "input_tokens": 1389,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.9900000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

Round 7: I sent \$2 and received a very good return (~\$5)

Round 8: As receiver, I got \$2.46 (from \$0.82 sent) and returned \$1

Round 7 showed excellent cooperation when I sent more. I'll send a similar amount to encourage that positive dynamic:

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Let me track what's happening:

Round 7: I sent \$2 and my earnings went from \$15.53 to \$18.53 = +\$3

So I sent \$2, earned back: \$2 sent + \$3 gained = I must have gotten \$5 back. That's excellent!

Round 8: As receiver, I got \$2.46 (from \$0.82 sent) and returned \$1

My earnings went from \$18.53 to \$19.99 = +\$1.46
That's correct: I received \$2.46 and returned \$1, so I kept \$1.46

Looking at the pattern:

- Round 7: Sender sent \$2, got great return (around \$5)
- Round 8: Sender sent \$0.82 (much less), got \$1 back from me

The sender seems to have reduced their send significantly in Round 8 after doing well in Round 7. This might be due to communication noise or a change in strategy.

I should send a moderate amount to encourage continued cooperation. Given that when I sent \$2 in Round 7 it seemed to work well, I could try that again or go slightly higher to signal trust.

I'll send \$2.5 to show willingness to cooperate and encourage reciprocity.

```
Usage: {  
  "input_tokens": 1356,  
  "output_tokens": 401,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:58:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is unfortunate as it breaks the cooperative pattern, but I have no choice in this situation.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0.

I can only return between 0 and 0, which means I can only return \$0.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust receivers based on their experience
2. They had a bad experience in a previous round
3. They're being strategic/cautious

Since I received \$0, I have nothing to return. The only possible response is to return \$0.

My total earnings are \$7.4 across all rounds, but that doesn't affect what I can do this round - I simply have no funds from the sender to return.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, these decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less than Round 2 to reflect the reduced trust level.

```
{'send': 1.50}
```

reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: $(\$5 - \$2.00 + \text{returned}) + \$2.58 = \$7.40$
- $\$3.00 + \text{returned} + \$2.58 = \$7.40$
- returned = \$1.82

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining

...

usage:

```
{
  "input_tokens": 931,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.9.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent \$0.65 (13% of endowment) - showing limited trust
- I received \$1.95

The cooperation level has been declining, but I want to maintain some reciprocity to potentially rebuild trust. The sender did show some willingness to cooperate, even if minimal.

I'll return approximately half of what I received, similar to my approach in Round 1. This ensures both of us gain something:

- Sender would get: $\$5 - \$0.65 + \$1.00 = \5.35
- I would get: $\$1.95 - \$1.00 = \$0.95$

This rewards their small gesture of trust while we both profit modestly.

```
{'return': 1.00}
```

reasoning:

Let me analyze this situation:

The sender sent \$0.65, which is 13% of their endowment. This is quite low compared to previous rounds where I've seen 37.2% and I sent 40%.

I received \$1.95 (the multiplication by 3).

Looking at the pattern:

- Round 1: I received \$5.58, returned \$3.00 (54%)
- Round 2: I sent \$2.00 (40%), but got low returns
- Round 3: Sender sent \$0, I returned \$0
- Round 4: I sent \$1.50
- Round 5: Now I received \$1.95 (sender sent \$0.65)

The cooperation has been declining. The sender only sent 13% of their endowment this round, showing limited trust.

However, I should consider:

1. There's communication noise - maybe they intended to send more
2. Reciprocity can rebuild trust
3. Being too punishing could end cooperation entirely

A fair approach would be to return roughly half of what I received, similar to my Round 1 strategy. This would give the sender a small profit for their trust while I also gain.

If I return \$1.00:

- Sender ends with: $\$5 - \$0.65 + \$1.00 = \5.35 (profit of \$0.35)
- I end with: $\$1.95 - \$1.00 = \$0.95$

This maintains some reciprocity while acknowledging the lower initial trust.

Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown, but still be cooperative.

Let me return \$0.90 (about 46% of what I received):

...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.850000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my experience:

****As Receiver:**** I've been fairly reciprocal (Round 1: returned \$3/\$5.58; Round 5: returned \$1/\$1.95)

****As Sender:****

- Round 2: Sent \$2.00, received \$1.82 back (decent reciprocity)
- Round 4: Sent \$1.50, received \$0 back (no reciprocity)

There's been inconsistency, possibly due to communication noise mentioned in the instructions. However, in Round 5, I showed fair reciprocity, which might encourage cooperation.

I'll send a moderate amount that balances trust with caution:

```
{'send': 1.75}
```

reasoning:

Let me review the rounds so far:

Round 1: I was receiver, sender sent \$1.86, I received \$5.58, I returned \$3.00

Round 2: I was sender, sent \$2.00, receiver returned some amount (I ended with \$7.4 total, so started with \$2.58, sent \$2.00, got \$5 + return - \$2 = \$3 + return. So \$7.4 - \$2.58 = \$4.82 gained. Started with \$5, sent \$2, so I kept \$3 + return. Return must have been \$1.82)

Round 3: I was receiver, sender sent \$0, I returned \$0. My total stayed at \$7.4

Round 4: I was sender, sent \$1.50, started with \$7.4, so I have \$5 from this round, sent \$1.50, so kept \$3.50 + return. Total is now \$10.9. So \$10.9 - \$7.4 = \$3.5, which means return was \$0. Hmm.

Round 5: I was receiver, sender sent \$0.65, I received \$1.95, I returned \$1.00. Started with \$10.9, gained \$0.95, so total is \$11.85.

So the pattern:

- Round 1: I received and reciprocated well
- Round 2: I sent \$2.00, got back \$1.82 (reasonable)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.50, got back \$0 (no reciprocation!)
- Round 5: I received \$0.65*3=\$1.95, returned \$1.00 (fair reciprocation)

So when I was receiver (Rounds 1, 3, 5), I was fairly generous.

When I was sender (Rounds 2, 4):

- Round 2 ...

usage:

```
{
  "input_tokens": 1360,
  "output_tokens": 837,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.100000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is disappointing after I sent a moderate amount in the previous round, but I have no choice.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is disappointing as I sent \$1.75 in Round 6, hoping to rebuild cooperation. But if the sender sent \$0, there's nothing I can do.

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.100000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$2.00 ? received \$1.82 (modest return)
- Round 4: Sent \$1.50 ? received \$0
- Round 6: Sent \$1.75 ? received \$0

The last two times I've sent money as sender, I received nothing back. This pattern suggests either low reciprocity or significant communication noise affecting the game.

While I've tried to maintain cooperation, the results when I'm the sender have been poor. I need to be more cautious.

I'll send a small amount to signal continued willingness to cooperate without risking much:

```
{'send': 0.75}
```

reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.00, received \$1.82 back
- Round 4: Sent \$1.50, received \$0 back
- Round 6: Sent \$1.75, received \$0 back (since Round 7 the sender sent me \$0)

As receiver:

- Round 1: Received \$5.58, returned \$3.00
- Round 3: Received \$0, returned \$0
- Round 5: Received \$1.95, returned \$1.00
- Round 7: Received \$0, returned \$0

The pattern shows declining cooperation overall. When I've been the sender recently (Round 4 and 6), I've gotten nothing back. This could be due to communication noise, or it could be that the receivers are not reciprocating.

My total earnings are \$15.10, which is averaging about \$1.89 per round across 8 rounds. This is below the \$5 I'd get by just keeping everything.

Given the pattern of low reciprocity when I'm the sender, I should be more cautious. However, completely stopping cooperation might eliminate any chance of future gains.

I'll send a small amount to test if there's any willingness to cooperate, but protect most of my endowment.

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 467,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is the third consecutive time I've received \$0 as receiver. The pattern of cooperation has completely broken down.

```
{'return': 0}
```

Reasoning:

The sender sent \$0 again, so I received \$0 and have nothing to return.

This is the third time in a row that when I'm the receiver, the sender has sent me \$0:

- Round 3: Sender sent \$0
- Round 7: Sender sent \$0
- Round 9: Sender sent \$0

This is frustrating, especially since:

- In Round 1, I received \$5.58 and returned \$3.00 (generous)

- In Round 5, I received \$1.95 and returned \$1.00 (fair)

Meanwhile, when I've been the sender:

- Round 2: Sent \$2.00, got \$1.82 back
- Round 4: Sent \$1.50, got \$0 back
- Round 6: Sent \$1.75, got \$0 back
- Round 8: Sent \$0.75, waiting to see what happens

The communication noise could be causing significant issues. Perhaps what I send/return is being perceived differently by others.

But regardless, I have \$0 to work with, so I must return \$0.

```
Usage: {
  "input_tokens": 1455,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 33.8,
"Agent_2": 48.2

}

Visible balances after round: {

"Agent_1": 23.39,
"Agent_2": 25.5

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.4.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

Round 1: I received and returned generously (54% of what I received)

Round 2: I sent \$2.00, but the receiver returned less generously (~30%)

Round 3: The sender sent \$0 to me

There's been declining cooperation. However, given the note about communication noise, the decisions might have been influenced by misperceptions.

I'll send a moderate amount to signal continued willingness to cooperate, but slightly less than Round 2 to reflect the reduced trust level.

{'send': 1.50}

reasoning:

Let me review what happened:

Round 1: I was RECEIVER

- Sender sent \$1.86, I received \$5.58
- I returned \$3.00
- My earnings: \$2.58

Round 2: I was SENDER

- I sent \$2.00 (which would become \$6.00 for receiver)
- My total earnings after round 2 are \$7.40
- So I started Round 2 with \$2.58 and ended with \$7.40
- That means: (\$5 - \$2.00 + returned) + \$2.58 = \$7.40
- \$3.00 + returned + \$2.58 = \$7.40
- returned = \$1.82

So the receiver returned \$1.82 on \$6.00 received (about 30%)

Round 3: I was RECEIVER

- Sender sent \$0, I received \$0
- I returned \$0 (couldn't return anything)
- My earnings stayed at \$7.40

Round 4: I am SENDER now

- My total earnings are \$7.40

The pattern so far:

- Round 1: Sender sent 37.2%, I returned about 54%
- Round 2: I sent 40%, receiver returned about 30%
- Round 3: Sender sent 0%, I returned 0% (had no choice)

The receiver in Round 2 was less generous than I was in Round 1. They only returned 30% of what they received, whereas I returned 54%.

In Round 3, the sender sent nothing, which could be:

1. A strategic decision to defect
2. A response to something (maybe they saw the Round 2 receiver's low return?)
3. Just testing

Now I need to decide what to do. Given:

- There's been some cooperation but declining

...

usage:

```
{
  "input_tokens": 931,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.9.

This round, The sender sent you \$0.65, so you received \$1.9500000000000002. How much do you return to The sender? (0-1.9500000000000002)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent \$0.65 (13% of endowment) - showing limited trust
- I received \$1.95

The cooperation level has been declining, but I want to maintain some reciprocity to potentially rebuild trust. The sender did show some willingness to cooperate, even if minimal.

I'll return approximately half of what I received, similar to my approach in Round 1. This ensures both of us gain something:

- Sender would get: $\$5 - \$0.65 + \$1.00 = \5.35
- I would get: $\$1.95 - \$1.00 = \$0.95$

This rewards their small gesture of trust while we both profit modestly.

{'return': 1.00}

reasoning:

Let me analyze this situation:

The sender sent \$0.65, which is 13% of their endowment. This is quite low compared to previous rounds where I've seen 37.2% and I sent 40%.

I received \$1.95 (the multiplication by 3).

Looking at the pattern:

- Round 1: I received \$5.58, returned \$3.00 (54%)
- Round 2: I sent \$2.00 (40%), but got low returns

- Round 3: Sender sent \$0, I returned \$0
- Round 4: I sent \$1.50
- Round 5: Now I received \$1.95 (sender sent \$0.65)

The cooperation has been declining. The sender only sent 13% of their endowment this round, showing limited trust.

However, I should consider:

1. There's communication noise - maybe they intended to send more
2. Reciprocity can rebuild trust
3. Being too punishing could end cooperation entirely

A fair approach would be to return roughly half of what I received, similar to my Round 1 strategy. This would give the sender a small profit for their trust while I also gain.

If I return \$1.00:

- Sender ends with: $\$5 - \$0.65 + \$1.00 = \5.35 (profit of \$0.35)
- I end with: $\$1.95 - \$1.00 = \$0.95$

This maintains some reciprocity while acknowledging the lower initial trust.

Alternatively, I could return a bit less (around 40-45%) to reflect the lower trust shown, but still be cooperative.

Let me return \$0.90 (about 46% of what I received):

...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.850000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience:

****As Receiver:**** I've been fairly reciprocal (Round 1: returned \$3/\$5.58; Round 5: returned \$1/\$1.95)

****As Sender:****

- Round 2: Sent \$2.00, received \$1.82 back (decent reciprocity)
- Round 4: Sent \$1.50, received \$0 back (no reciprocity)

There's been inconsistency, possibly due to communication noise mentioned in the instructions. However, in Round 5, I showed fair reciprocity, which might encourage cooperation.

I'll send a moderate amount that balances trust with caution:

```
{'send': 1.75}
```

reasoning:

Let me review the rounds so far:

Round 1: I was receiver, sender sent \$1.86, I received \$5.58, I returned \$3.00

Round 2: I was sender, sent \$2.00, receiver returned some amount (I ended with \$7.4 total, so started with \$2.58, sent \$2.00, got \$5 + return - \$2 = \$3 + return. So \$7.4 - \$2.58 = \$4.82 gained. Started with \$5, sent \$2, so I kept \$3 + return. Return must have been \$1.82)

Round 3: I was receiver, sender sent \$0, I returned \$0. My total stayed at \$7.4

Round 4: I was sender, sent \$1.50, started with \$7.4, so I have \$5 from this round, sent \$1.50, so kept \$3.50 + return. Total is now \$10.9. So \$10.9 - \$7.4 = \$3.5, which means return was \$0. Hmm.

Round 5: I was receiver, sender sent \$0.65, I received \$1.95, I returned \$1.00. Started with \$10.9, gained \$0.95, so total is \$11.85.

So the pattern:

- Round 1: I received and reciprocated well
- Round 2: I sent \$2.00, got back \$1.82 (reasonable)
- Round 3: Received \$0, returned \$0
- Round 4: I sent \$1.50, got back \$0 (no reciprocation!)
- Round 5: I received $0.65 \times 3 = \$1.95$, returned \$1.00 (fair reciprocation)

So when I was receiver (Rounds 1, 3, 5), I was fairly generous.

When I was sender (Rounds 2, 4):

- Round 2 ...

usage:

```
{
  "input_tokens": 1360,
  "output_tokens": 837,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.100000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is disappointing after I sent a moderate amount in the previous round, but I have no choice.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0. I have nothing to return.

This is disappointing as I sent \$1.75 in Round 6, hoping to rebuild cooperation. But if the sender sent \$0, there's nothing I can do.

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.100000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$2.00 ? received \$1.82 (modest return)
- Round 4: Sent \$1.50 ? received \$0
- Round 6: Sent \$1.75 ? received \$0

The last two times I've sent money as sender, I received nothing back. This pattern suggests either low reciprocity or significant communication noise affecting the game.

While I've tried to maintain cooperation, the results when I'm the sender have been poor. I need to be more cautious.

I'll send a small amount to signal continued willingness to cooperate without risking much:

```
{'send': 0.75}
```

reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.00, received \$1.82 back
- Round 4: Sent \$1.50, received \$0 back
- Round 6: Sent \$1.75, received \$0 back (since Round 7 the sender sent me \$0)

As receiver:

- Round 1: Received \$5.58, returned \$3.00
- Round 3: Received \$0, returned \$0
- Round 5: Received \$1.95, returned \$1.00
- Round 7: Received \$0, returned \$0

The pattern shows declining cooperation overall. When I've been the sender recently (Round 4 and 6), I've gotten nothing back. This could be due to communication noise, or it could be that the receivers are not reciprocating.

My total earnings are \$15.10, which is averaging about \$1.89 per round across 8 rounds. This is below the \$5 I'd get by just keeping everything.

Given the pattern of low reciprocity when I'm the sender, I should be more cautious. However, completely stopping cooperation might eliminate any chance of future gains.

I'll send a small amount to test if there's any willingness to cooperate, but protect most of my endowment.

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 467,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0, I received \$0 and have nothing to return.

This is the third consecutive time I've received \$0 as receiver. The pattern of cooperation has completely broken down.

{'return': 0}

reasoning:

The sender sent \$0 again, so I received \$0 and have nothing to return.

This is the third time in a row that when I'm the receiver, the sender has sent me \$0:

- Round 3: Sender sent \$0
- Round 7: Sender sent \$0
- Round 9: Sender sent \$0

This is frustrating, especially since:

- In Round 1, I received \$5.58 and returned \$3.00 (generous)
- In Round 5, I received \$1.95 and returned \$1.00 (fair)

Meanwhile, when I've been the sender:

- Round 2: Sent \$2.00, got \$1.82 back
- Round 4: Sent \$1.50, got \$0 back
- Round 6: Sent \$1.75, got \$0 back
- Round 8: Sent \$0.75, waiting to see what happens

The communication noise could be causing significant issues. Perhaps what I send/return is being perceived differently by others.

But regardless, I have \$0 to work with, so I must return \$0.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 323,
}
```

```

    "reasoning_tokens": 0
  }
Message 14 role=user
  content:
    Round 10

    You are the SENDER this round. Your total earnings across all rounds are $20.5.
    How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at the complete pattern:

  **As Sender:**
  - Round 2: Sent $2.00 ? received $1.82
  - Round 4: Sent $1.50 ? received $0
  - Round 6: Sent $1.75 ? received $0
  - Round 8: Sent $0.75 ? received $0 (based on unchanged earnings)

  **As Receiver:**
  - Last three rounds: Received $0 each time

  Cooperation has completely collapsed. I've tried sending various amounts ($2.00, $1.75, $1.50, $0.75), but the last three rounds as sender yielded nothing. Meanwhile, as receiver, I've gotten $0 three consecutive times.

  This appears to be the final round (Round 10). Given the consistent pattern of non-cooperation, sending anything would be irrational.

  {'send': 0}
Reasoning:
  Looking at my history as SENDER:
  - Round 2: Sent $2.00 ? received $1.82 back
  - Round 4: Sent $1.50 ? received $0 back
  - Round 6: Sent $1.75 ? received $0 back
  - Round 8: Sent $0.75 ? received $0 back (presumably, since my earnings are $20.5)

  My total earnings are $20.5, which is exactly what they were after Round 9, suggesting I received nothing back in Round 8.

  As RECEIVER:
  - Round 1: Received $5.58, returned $3.00
  - Round 3: Received $0, returned $0
  - Round 5: Received $1.95, returned $1.00
  - Round 7: Received $0, returned $0
  - Round 9: Received $0, returned $0

  I've been fairly reciprocal as a receiver when I get something, but as a sender, I've gotten almost nothing back over the last 3 rounds.

  This is Round 10, possibly the final round. Given the complete breakdown in cooperation, and that senders in my position have repeatedly sent $0 to me, the rational move is to keep my money.

  The game has devolved into non-cooperation. Sending anything at this point would be irrational given the pattern.
  Usage: {
    "input_tokens": 1445,
    "output_tokens": 532,
    "reasoning_tokens": 0
  }

GAME - Agent_1 - Receiver - dyad_1
  Timestamp: 2026-08-12T09:58:40
  Model: anthropic/claude-sonnet-4.5
  Temperature: 0.8
  Metadata: {
    "agent_type": "standard",
    "dyad_id": "dyad_1",
    "move_index": 1,
    "opponent_id": "Agent_2",
    "pairing": {

```

```

"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.760000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent me \$0, so I received \$0
- There's nothing for me to return since I didn't receive anything

This is a notable shift from the cooperative pattern we'd established. However, I recall that

he instructions mentioned there may be "communication noise" that could affect the amounts. Perhaps this wasn't intentional.

Since I received \$0, I can only return:

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0. Since there's nothing to return, I can only return \$0. This is a bit concerning - after we had been cooperating, the sender chose not to send anything this round.

However, I should note the instructions mentioned there may be communication noise - amounts I see may differ from what was actually sent. So it's possible the sender meant to send something but noise reduced it to 0, or perhaps they sent a small amount that got rounded down.

Since I received \$0, I can only return between \$0 and \$0.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 879,
  "output_tokens": 248,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.7600000000000002.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent pattern:

Round 3: I sent \$2.5 and my earnings only increased by \$2.5 (meaning I got \$0 back)

Round 4: As receiver, I got \$0 from the sender

Cooperation seems to have broken down. However, the instructions mentioned "communication noise" - these zeros might not reflect actual intentions.

I'll send a small amount to test if we can rebuild cooperation while protecting most of my funds:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me think about what's happened:

Round 1: I was SENDER, sent \$2, got a good return (~\$3)

Round 2: I was RECEIVER, got \$0.9 (became \$2.7), returned \$1.5

Round 3: I was SENDER, sent \$2.5, but then...

Round 4: I was RECEIVER, got \$0, returned \$0

Wait, in Round 4, the sender sent me \$0. But my earnings stayed at \$9.76 from Round 3.

Let me trace this more carefully:

- After Round 2, my total was \$7.26
- In Round 3, I was SENDER and sent \$2.5
- After Round 3, my total is \$9.76

So in Round 3:

- I started with \$5, sent \$2.5
- I kept \$2.5

- I should have received something back
- My total went from \$7.26 to \$9.76, an increase of \$2.5
- This means I received \$0 back from the receiver in Round 3

Then in Round 4:

- I received \$0 (sender sent \$0)
- My total stayed at \$9.76

So the pattern is:

- Round 1: I sent \$2, got ~\$3 back (good cooperation)
- Round 2: As receiver, I got \$0.9, returned \$1.5 (I was generous)
- Round 3: I sent \$2.5, got \$0 back (receiver defected)
- Round 4: Sender sent me \$0 (they stopped cooperating)

The game mentioned there could be "communication noise" - amounts might differ from what was actually sent. But it seems like cooperation has broken down.

Given this pattern of defection, I need to be more cautious. However, completely stopping cooperation means we both lose out on potential gains.

...

usage:

```
{
  "input_tokens": 1029,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.850000000000001.

This round, The sender sent you \$0.46, so you received \$1.3800000000000001. How much do you return to The sender? (0-1.3800000000000001)

Message 7 role=assistant

content:

Looking at this round:

- The sender sent \$0.46, which became \$1.38 for me
- This is a small send, possibly testing cooperation after recent issues
- I want to encourage rebuilding trust by being fair

I'll return a reasonable amount that allows both of us to profit from this exchange:

```
'''json
{"return": 0.7}
'''
```

reasoning:

Let me analyze this situation:

Round 5: I sent \$1 as a sender (presumably to this receiver)

Round 6: As receiver, I'm now getting \$0.46 which became \$1.38

The sender is sending a relatively small amount (\$0.46), possibly testing the waters after the breakdown in cooperation.

Looking at my earnings jump from \$9.76 to \$14.85, that's an increase of about \$5.09. That suggests in Round 5, I received a decent return - maybe around \$5 back after sending \$1.

Now the sender is being cautious with only \$0.46. I should return a fair amount to encourage rebuilding trust. Since they sent \$0.46 and I received \$1.38, I should return enough that they profit slightly.

If I return around \$0.7-\$0.9, the sender would:

- Started with \$5
- Sent \$0.46, has \$4.54
- Gets back ~\$0.7-\$0.9
- Ends with ~\$5.24-\$5.44 (profit of \$0.24-\$0.44)

I would keep:

- \$1.38 - \$0.8 = \$0.58 profit

Let me return about \$0.7 to be fair and signal cooperation.

usage:

```
{
  "input_tokens": 1225,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.530000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at recent trends:

Round 5: I sent \$1 and got back about \$1.09 (modest but positive)

Round 6: As receiver, I got \$1.38 (from \$0.46 sent) and returned \$0.7 fairly

Cooperation seems to be rebuilding after the breakdown in Rounds 3-4. I'll send a moderate amount to encourage this positive trend while being cautious:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track the history:

Round 1 (SENDER): I sent \$2

Round 2 (RECEIVER): I received \$2.7 (from \$0.9 sent), returned \$1.5

Round 3 (SENDER): I sent \$2.5

Round 4 (RECEIVER): I received \$0 (from \$0 sent), returned \$0

Round 5 (SENDER): I sent \$1

Round 6 (RECEIVER): I received \$1.38 (from \$0.46 sent), returned \$0.7

My total earnings are \$15.53.

Let me trace what I should have earned:

- Round 1: Started with \$5, sent \$2, kept \$3. Need to know what was returned.
- Round 2: As receiver, got \$2.7, returned \$1.5, kept \$1.2
- Round 3: Started with \$5, sent \$2.5, kept \$2.5. Need to know what was returned.
- Round 4: As receiver, got \$0, returned \$0, kept \$0
- Round 5: Started with \$5, sent \$1, kept \$4. Need to know what was returned.
- Round 6: As receiver, got \$1.38, returned \$0.7, kept \$0.68

Let me work backwards from \$15.53:

- After Round 6: earned \$0.68, so before Round 6 I had $\$15.53 - \$0.68 = \$14.85$ (matches what was stated)
- After Round 5: Let's say I got back X from Round 5. Before Round 5 I had $\$14.85 - \$4 - X = \$10.85 - X$
- After Round 4: earned \$0, so before Round 4 I had $\$10.85 - X$
- After Round 3: If I got back Y from Round 3, before Round 3 I had $\$10.85 - X - \$2.5 - Y = \$8.35 - X - Y$
- After Round 2: earned \$1.2, so before Round 2 I had $\$7.15 - X - Y$
- After Round 1: If I got back Z from Round 1, before Round 1 I had $\$4.15 - X - Y - Z$

Since I started with \$0, the returns must sum to \$4.15: $X + Y + Z = \$4.15$

From the pattern of each round:

- Round 1: Sent \$2 which became \$6, earnings were \$6.06, so $Z = \$3.06$
- Round 3: Sent \$2.5 which became \$7.5, earnings were \$9.76, and I had \$7.26 before this round, meaning $Y = 0$
- Round 5: Sent \$1 which became \$3, and I had \$14.85 after this round

, so $X = \$0.41$

Let me recalculate systematically. Round 1 as sender: started with \$5, sent \$2, kept \$3, ended with \$6.06, so received back \$3.06. Round 2 as receiver: got \$2.7, returned \$1.5, kept \$1.2, total became \$7.26. Round 3 as sender: started with \$5, sent \$2.5, kept \$2.5, ended with \$9.76, so received back \$0. Round 4 as receiver: received \$0.

In Round 5 as sender, I started with \$5, sent \$1, kept \$4. Before Round 6 I had \$14.85, and I kept \$0.68 in that round, so after Round 5 I must have had \$14.85, meaning I received back \$1.09. Round 6 as receiver: received \$1.38, returned \$0.7, kept \$0.68, bringing my total to \$15.53....

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 1594,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$18.53.

This round, The sender sent you \$0.82, so you received \$2.46. How much do you return to The sender? (0-2.46)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$0.82, which became \$2.46 for me
- In Round 7, I sent \$2 but seems I got little/nothing back, which is disappointing
- However, continuing a cycle of non-cooperation hurts both of us

I'll return a fair amount to signal willingness to cooperate, but be somewhat cautious given recent history:

```
'''json
{"return": 1}
'''
```

reasoning:

Looking at the pattern:

Round 7: I sent \$2 and my earnings went from \$15.53 to \$18.53 = \$3 increase

This means: $\$5 - \$2 + \text{returned amount} = \3

So: $\$3 + \text{returned amount} = \3

Therefore: returned amount ? \$0

So I sent \$2, and apparently got \$0 back (or close to it).

Round 8 (now): As receiver, I got \$0.82 which became \$2.46

This is frustrating. I sent \$2 last round and got nothing back. Now the sender only sends \$0.82.

However, I need to consider:

1. There's communication noise - maybe the sender intended more
2. I committed to being fair and cooperative
3. If I retaliate by returning nothing, we're in a downward spiral

Let me think about what's fair:

- The sender sent \$0.82 out of \$5, keeping \$4.18
- I received \$2.46
- If I return about half (~\$1.2), both of us profit from the exchange
- The sender would get: $\$5 - \$0.82 + \$1.2 = \5.38
- I would get: $\$2.46 - \$1.2 = \$1.26$

That seems reasonable. But given they returned \$0 last round, maybe I should return less?

Actually, let me recalculate Round 7:

- I started Round 7 at \$15.53
- I sent \$2
- I ended Round 7 at \$18.53
- So I gained \$3

If I sent \$2, I should have: $\$5 - \$2 = \$3$ remaining, plus whatever was returned.
 $\$3 + \text{returned} = \3
 $\text{returned} = \$0$

Yeah, so they returned nothing to me in Round 7 when I sent \$2.

...

usage:

```
{
  "input_tokens": 1389,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.9900000000000002.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the recent pattern:

Round 7: I sent \$2 and received a very good return (~\$5)

Round 8: As receiver, I got \$2.46 (from \$0.82 sent) and returned \$1

Round 7 showed excellent cooperation when I sent more. I'll send a similar amount to encourage that positive dynamic:

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me track what's happening:

Round 7: I sent \$2 and my earnings went from \$15.53 to \$18.53 = +\$3

So I sent \$2, earned back: \$2 sent + \$3 gained = I must have gotten \$5 back. That's excellent!

Round 8: As receiver, I got \$2.46 (from \$0.82 sent) and returned \$1

My earnings went from \$18.53 to \$19.99 = +\$1.46

That's correct: I received \$2.46 and returned \$1, so I kept \$1.46

Looking at the pattern:

- Round 7: Sender sent \$2, got great return (around \$5)
- Round 8: Sender sent \$0.82 (much less), got \$1 back from me

The sender seems to have reduced their send significantly in Round 8 after doing well in Round 7. This might be due to communication noise or a change in strategy.

I should send a moderate amount to encourage continued cooperation. Given that when I sent \$2 in Round 7 it seemed to work well, I could try that again or go slightly higher to signal trust.

I'll send \$2.5 to show willingness to cooperate and encourage reciprocity.

usage:

```
{
  "input_tokens": 1356,
  "output_tokens": 401,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$23.39.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this round:

- The sender sent me \$0, so I received \$0
- This is the second time receiving \$0 (also happened in Round 4)

This is disappointing after I sent \$2.5 in Round 9, but there's nothing I can return since I didn't receive anything. The "communication noise" mentioned in the instructions might be affecting these transactions.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0. This is the second time this has happened (also in Round 4).

Since I received nothing, I can only return:

```
'''json
{"return": 0}
'''
```

This is disappointing after what seemed like rebuilding cooperation in Round 9 where I sent \$2.5. The instructions mentioned "communication noise" so this might not be intentional. Either way, I have to return 0 since I received 0.

```
Usage: {
  "input_tokens": 1358,
  "output_tokens": 215,
  "reasoning_tokens": 0
}
```